ZKTrustLLM-Agents Level 4 Technical Documentation

Steps 77-93: proof-governed MCP/A2A coordination, live RTP media telemetry, DTLS-wrapped RTP validation, impairment testing, and namespace-based media-plane evaluation

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1. Executive Summary

This report documents the Level 4 ZKTrustLLM-Agents project progress. The work has evolved from proof-governed MCP/A2A agent coordination into live media-plane validation. The project now contains measured evidence for semi-live MCP/A2A control-plane latency, multi-agent scaling, plain RTP media delivery, DTLS-wrapped RTP protection, loopback impairment, and Linux network namespace impairment. The result is a stronger research foundation for supervisor review and future journal-level evaluation.

2. Research Architecture

Plane	Role	Implemented Evidence
Media plane	Carries RTP/H.264 traffic and measures packet delivery behaviour.	Step 89 plain RTP, Step 90 DTLS-wrapped RTP, Step 91 loopback impairment, Step 93 namespace impairment.
Control plane	Runs MCP/A2A coordination and compact reference exchange between agents.	Step 86 semi-live MCP/A2A timings and Step 87 multi-agent scaling.
Trust plane	Uses blockchain-authenticated state, reference-bound decisions, and proof-governed policy admissibility.	AUTH_V2.3, reference bundle verification, negative-security rejection, and policy binding.
Evidence plane	Keeps reproducible artifacts, CSV logs, JSON summaries, figures, and report material.	ARTIFACTS.md, results folders, paper tables, report PDFs, and generated figures.

3. Step-by-Step Project Progress

Step	Main Contribution	Research Meaning
77	A2A reference-aware coordination	Introduced compact inter-agent reference exchange.
79	Proper MCP Level 4 context server	Structured tool access to authenticated blockchain state.
80	MCP/A2A KPI analysis	Defined measurable control-plane KPIs.
81	AUTH_V2.3 reference-bound proof	Bound proof-governed decisions to references and policy context.
82	Negative-security tests	Validated rejection of invalid policy or reference states.
83	Network KPI telemetry emulation	Created deterministic network KPI baseline.
84	Journal evaluation write-up	Converted prototype outputs into paper-ready evaluation material.
85	Benchmark result figures	Added visual result figures for agent scaling and utility.
86	Semi-live MCP/A2A control telemetry	Measured actual local MCP/A2A control-loop timings.
87	Multi-agent scaling telemetry	Measured performance as agent count increased from 5 to 25.
88	Technical report PDF through Step 87	Produced supervisor-ready progress report.
89	Live RTP media-plane telemetry	Measured actual RTP packets, bitrate, jitter, and loss.
90	DTLS-wrapped RTP validation	Added secured RTP tunnel/proxy validation.
91	Loopback network impairment	Compared plain RTP vs DTLS-RTP under tc netem impairment.
92	Technical report through Step 91	Updated report to include media-plane validation.
93	Linux network namespace impairment	Reproduced impairment matrix using two isolated network stacks.

4. Key Measured Results

4.1 Step 86 Semi-Live MCP/A2A Control-Plane Telemetry

Metric	Result
L4 raw-context average control response	72.793 ms
L4-ref MCP/A2A average control response	34.918 ms
L4-ref latency reduction vs raw-context	52.03%
L4-ref control-message reduction vs raw-context	97.91%
Interpretation	Compact reference exchange reduces control latency and message size while preserving proof-governed verification.

4.2 Step 87 Multi-Agent Scaling

Agents	Latency reduction vs raw	Message reduction vs raw	Throughput gain vs raw
5	53.14%	97.75%	113.51%
10	53.67%	97.76%	116.03%
15	58.50%	97.75%	139.83%
20	54.69%	97.74%	120.75%
25	52.40%	97.74%	110.02%

4.3 Step 89 Live Plain RTP Media-Plane Baseline

KPI	Result
Received packets	2328
Average bitrate kbps	833.605
Packet loss %	0
Average jitter ms	0.229169
P50 jitter ms	0.069357
Max jitter ms	4.16395

4.4 Step 90 DTLS-Wrapped RTP Media-Plane Validation

KPI	Result
Recovered RTP packets	769
Average bitrate kbps	102.17
Packet loss %	0
Average arrival-gap jitter ms	0.95001
P50 jitter ms	0.1185
Max jitter ms	33.3165

4.5 Step 91 Loopback Network Impairment

Profile	Mode	Packets	Loss %	Bitrate kbps	Avg jitter ms	P95 jitter ms
clean_baseline	plain_rtp	2328	0	833.605	10.7091	34.019
clean_baseline	dtls_rtp	769	0	102.17	0.926337	0.4245
delay_20ms	plain_rtp	2328	0	833.605	10.7031	33.846
delay_20ms	dtls_rtp	769	0	102.419	0.98316	0.5325
delay_20ms_jitter_5ms	plain_rtp	2328	28.6983	833.709	8.32631	24.453
delay_20ms_jitter_5ms	dtls_rtp	769	2.41117	102.419	10.2768	24.8545
delay_30ms_jitter_10ms_loss_1pct	plain_rtp	2300	28.5492	823.692	6.97549	19.719
delay_30ms_jitter_10ms_loss_1pct	dtls_rtp	737	13.7002	98.3696	17.4648	37.8075

4.6 Step 93 Linux Network Namespace Impairment

Profile	Mode	Packets	Loss %	Bitrate kbps	Avg jitter ms	P95 jitter ms
clean_baseline	plain_rtp_namespace	752	0	100.404	1.55151	0.399285
clean_baseline	dtls_rtp_namespace	692	0	92.7651	1.7021	0.657679
delay_20ms	plain_rtp_namespace	752	0	100.404	1.54312	0.405098
delay_20ms	dtls_rtp_namespace	691	0	92.7018	1.72985	1.31131
delay_20ms_jitter_5ms	plain_rtp_namespace	752	0	100.404	7.03223	18.9516
delay_20ms_jitter_5ms	dtls_rtp_namespace	691	0	92.7018	6.74515	20.9024
delay_30ms_jitter_10ms_loss_1pct	plain_rtp_namespace	744	0.932091	99.2202	12.3346	30.5731
delay_30ms_jitter_10ms_loss_1pct	dtls_rtp_namespace	684	0.869565	91.9613	12.0147	30.425

5. Result Figures

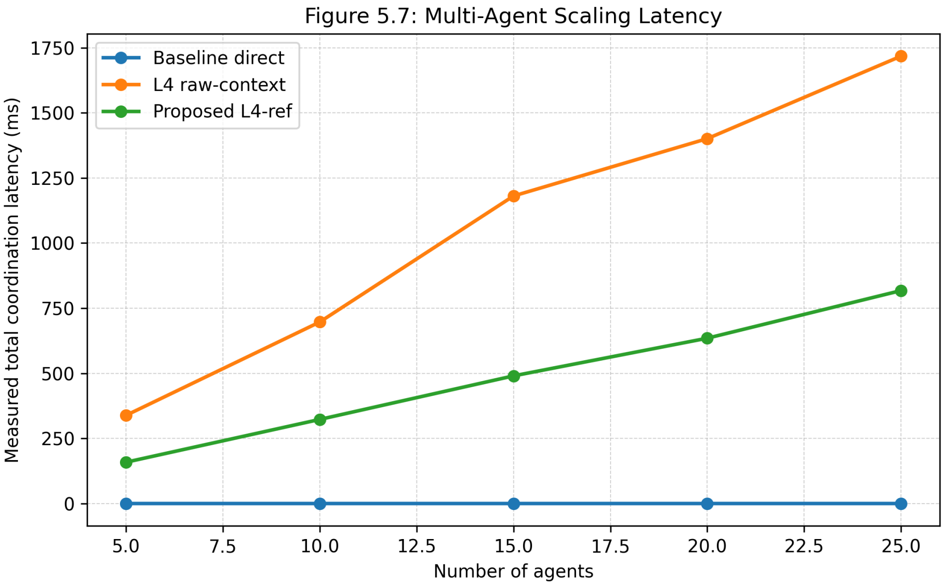

Figure 5.7 - Multi-agent coordination latency.

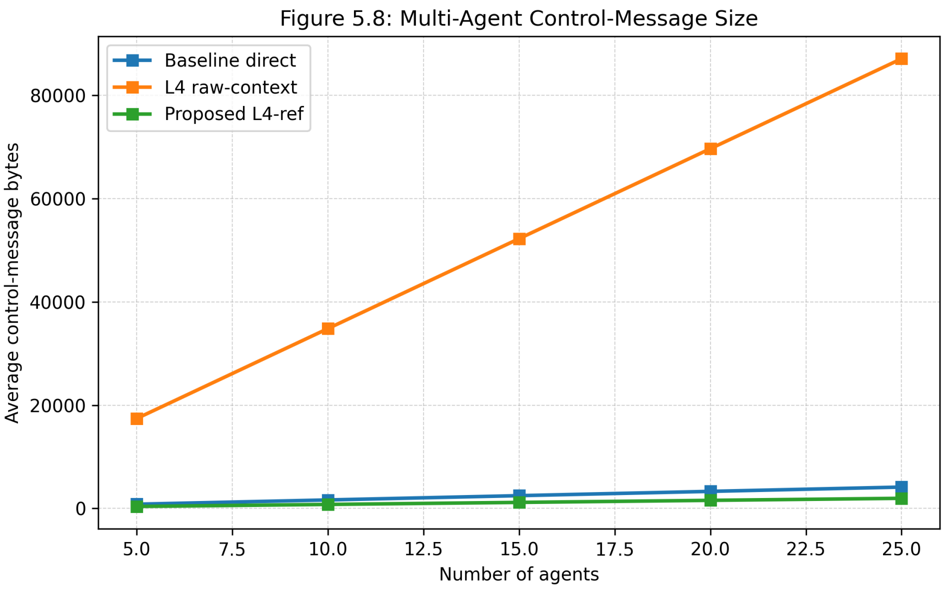

Figure 5.8 - Multi-agent control message bytes.

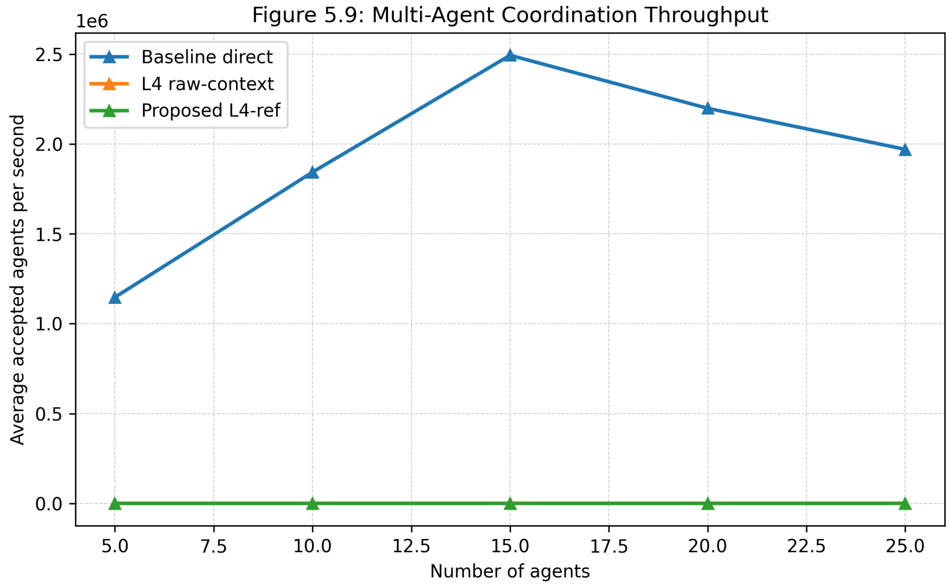

Figure 5.9 - Multi-agent throughput.

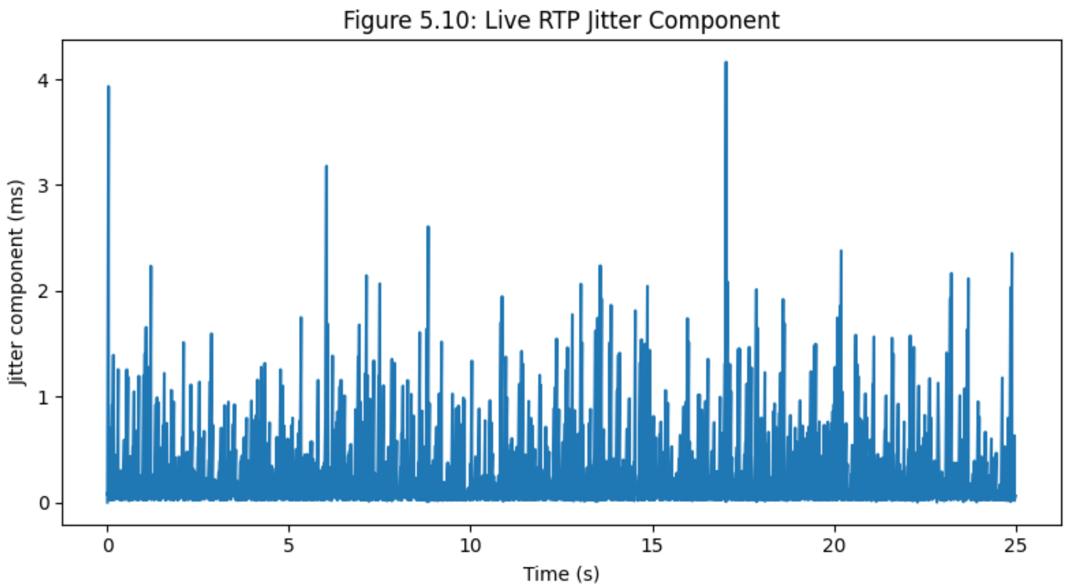

Figure 5.10 - Live RTP jitter.

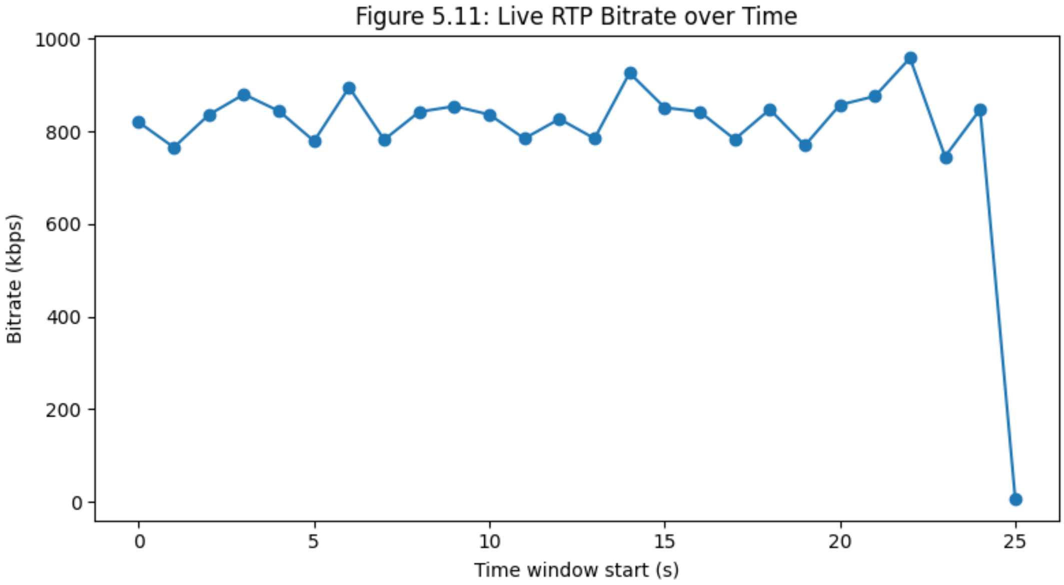

Figure 5.11 - Live RTP bitrate.

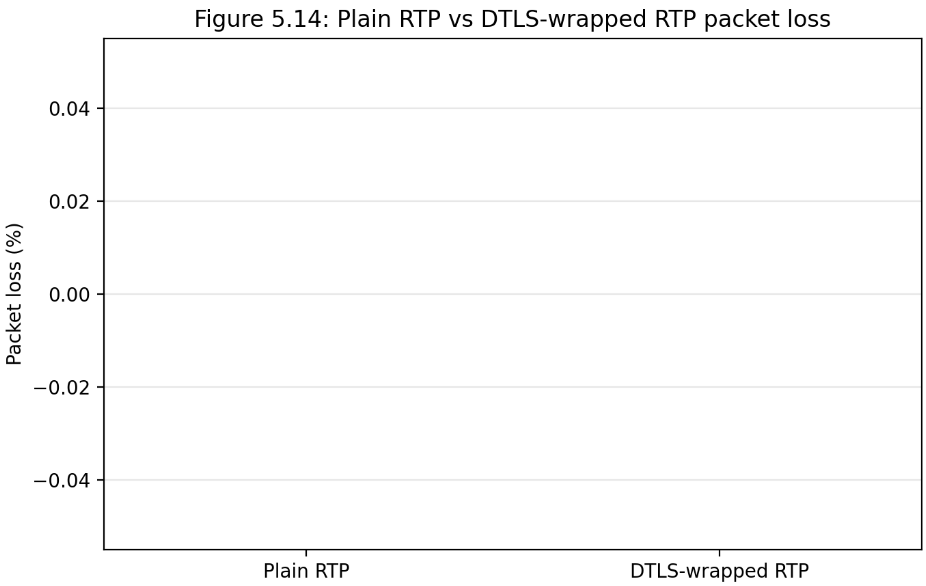

Figure 5.14 - Plain RTP vs DTLS-RTP packet loss.

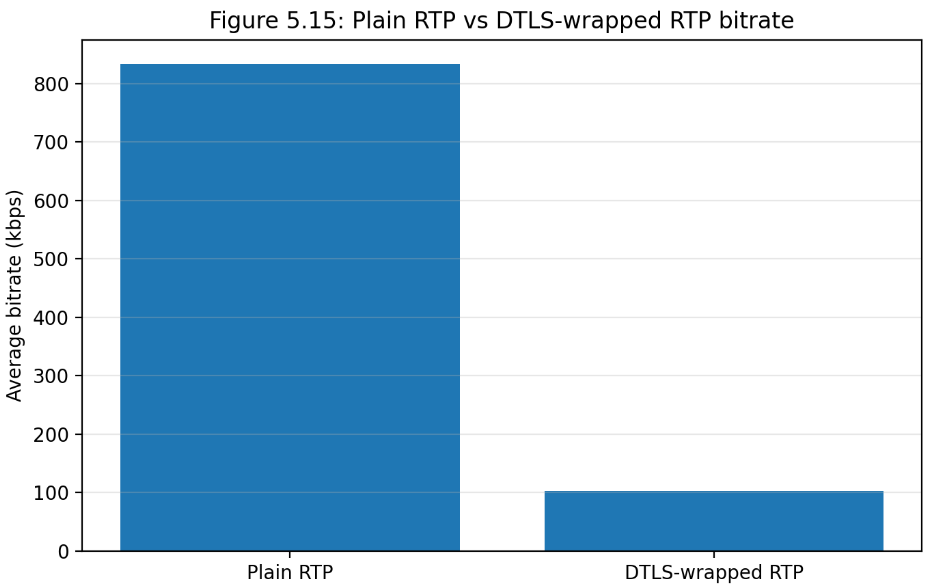

Figure 5.15 - Plain RTP vs DTLS-RTP bitrate.

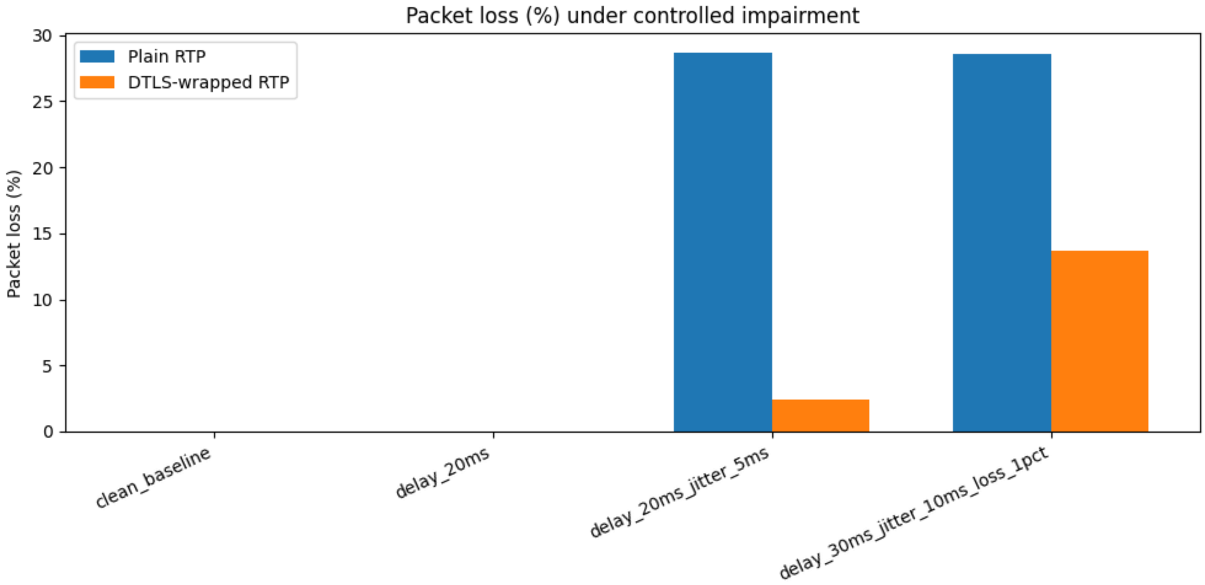

Figure 5.16 - Loopback impairment packet loss.

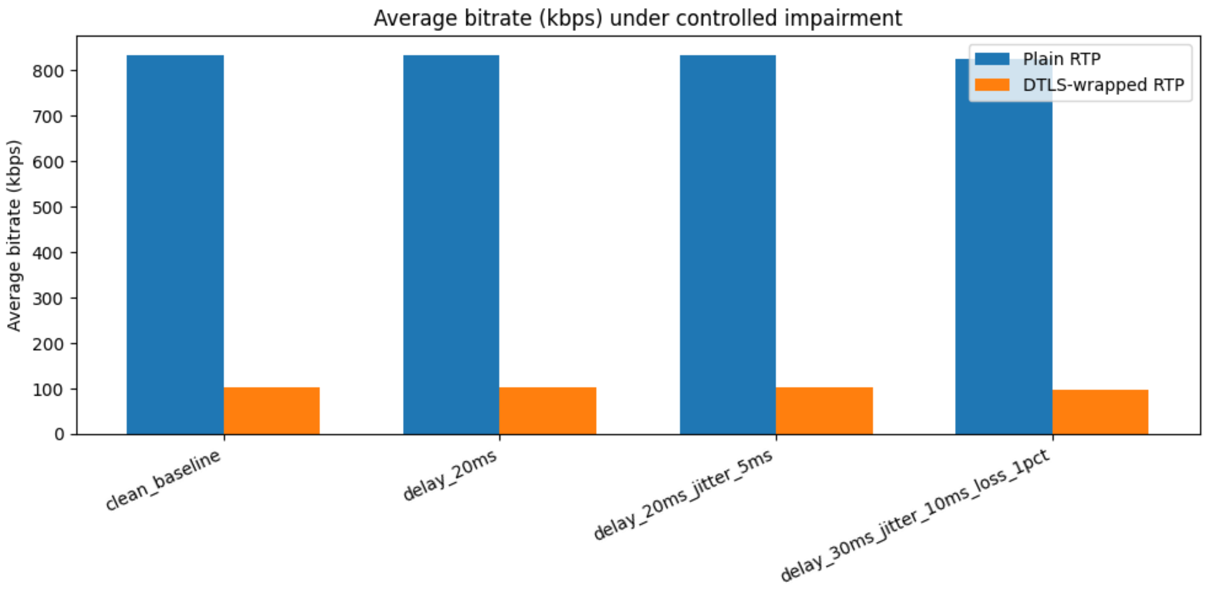

Figure 5.17 - Loopback impairment bitrate.

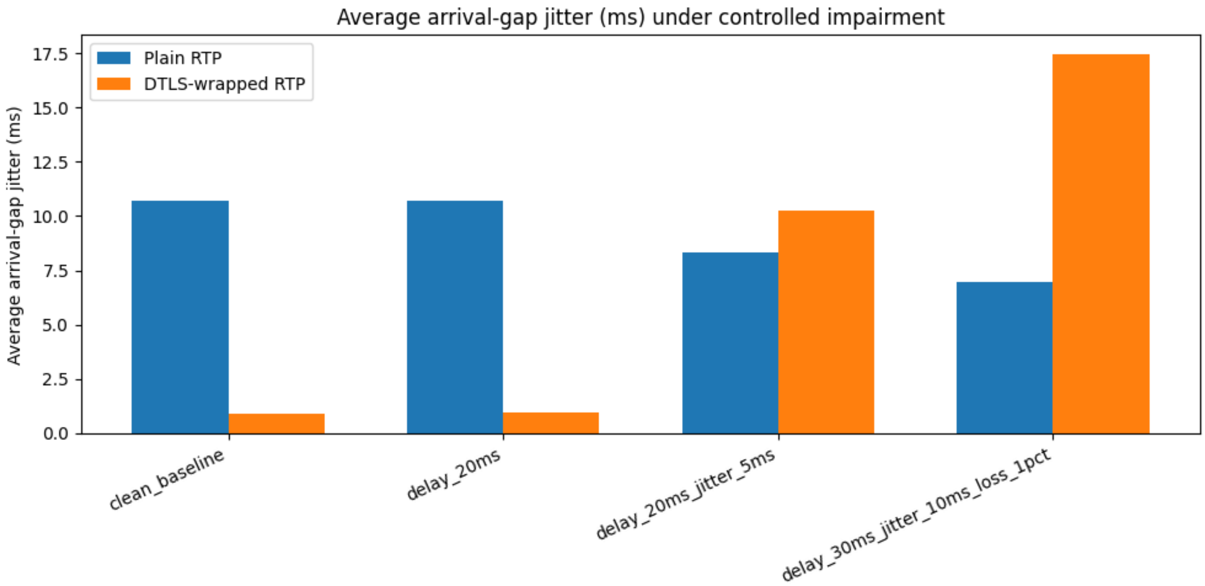

Figure 5.18 - Loopback impairment arrival-gap jitter.

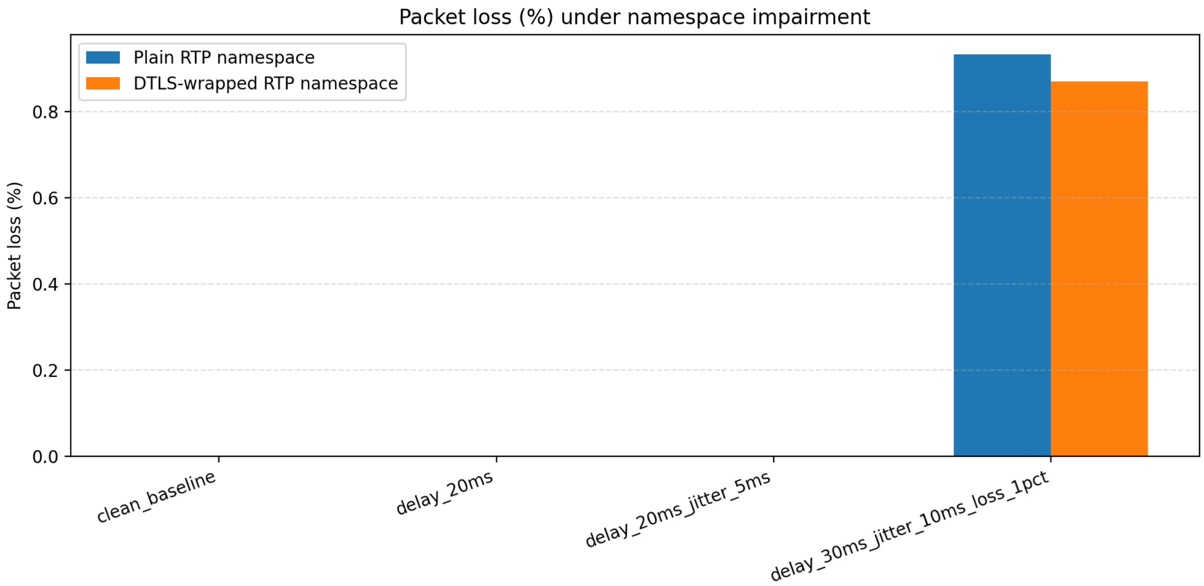

Figure 5.19 - Namespace packet loss.

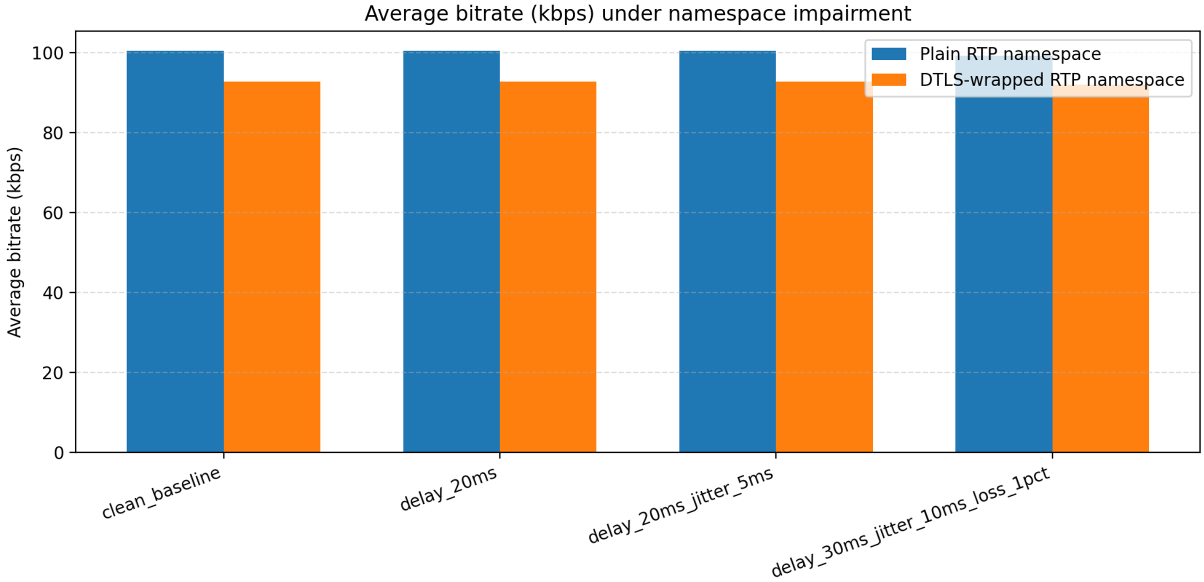

Figure 5.20 - Namespace bitrate.

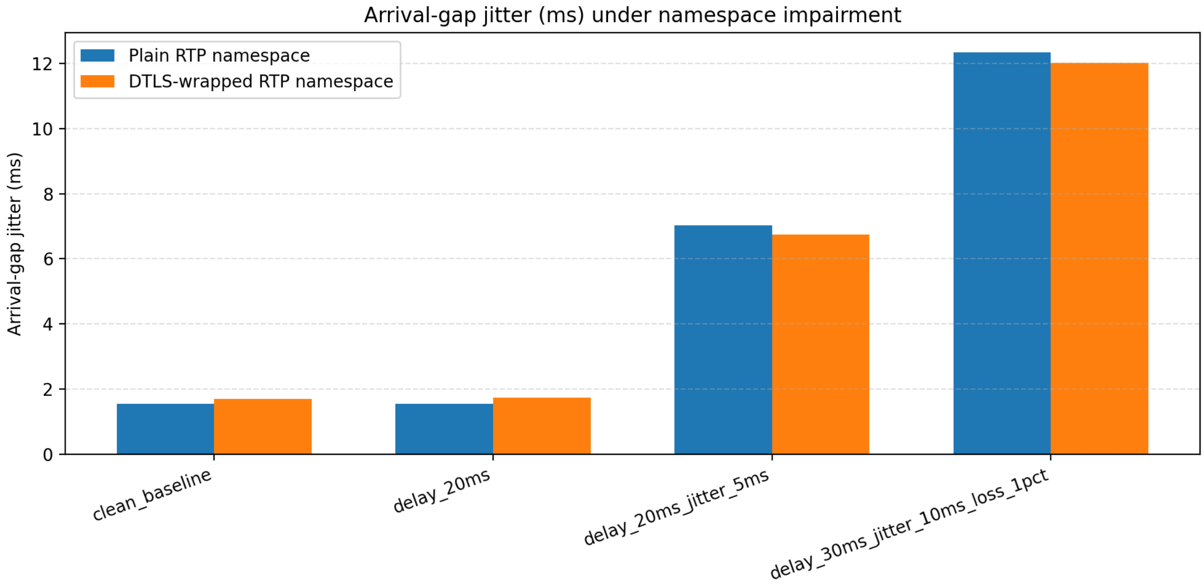

Figure 5.21 - Namespace arrival-gap jitter.

6. Supervisor Feedback Alignment

Supervisor/Research Need	Implemented Response	Evidence
Move beyond deterministic emulation.	Added semi-live MCP/A2A timings and live media-plane telemetry.	Steps 86, 89.
Evaluate MCP/A2A at Level 4.	Measured compact reference coordination against raw-context coordination.	Steps 86 and 87.
Connect control-plane work with media-plane behaviour.	Added RTP packet capture, packet loss, jitter, and bitrate metrics.	Steps 89 and 90.
Add secured media validation.	Built DTLS-wrapped RTP tunnel/proxy and measured recovered RTP.	Step 90.
Evaluate under network stress.	Used Linux tc netem for delay, jitter, and packet-loss impairment.	Step 91.
Improve realism beyond localhost.	Reproduced impairment matrix using two Linux network namespaces and veth.	Step 93.
Prepare journal-ready evaluation.	Generated result tables, figures, reports, and artifact mappings.	Steps 84, 88, 92, 94.

7. Artifact Map

Category	Main Paths
Control telemetry	results/l4_live_telemetry/
Multi-agent scaling	results/l4_multi_agent_scaling/
Plain RTP media	results/l4_live_rtp_media/
DTLS-RTP media	results/l4_dtls_rtp_media/
Loopback impairment	results/l4_network_impairment/
Namespace impairment	results/l4_namespace_impairment/
Technical reports	results/l4_report_pdf/ and docs/report/
Paper write-up	docs/paper/l4_mcp_a2a_evaluation_section.md and docs/paper/l4_results_tables.md
Artifact register	ARTIFACTS.md

8. Agentic AI Automation Roadmap From Here

Next Step	Goal	Expected Output
Step 95	Closed-loop policy-to-media experiment.	Trigger media impairment response from MCP/A2A policy output.
Step 96	Agentic AI control automation.	Automated agent reads telemetry, selects policy action, and logs decision.
Step 97	Supervisor feedback tracker.	Machine-readable feedback register linking comments to implementation evidence.
Step 98	Two-machine or testbed validation.	Repeat namespace results across two physical devices or 5GIC/6GIC testbed.
Step 99	Paper-ready result consolidation.	Clean tables, final figures, limitations, and journal evaluation section.
Step 100	Agentic L4 journal prototype.	Multi-agent L4 workflow with MCP/A2A, ZK proof governance, and secured media-plane response.

9. Current Research Claim

The Level 4 prototype demonstrates that proof-governed agentic coordination can combine MCP-based access to authenticated state, A2A compact reference exchange, reference-bound proof semantics, and secured RTP media-plane telemetry. The evaluation now includes control-plane latency, multi-agent scaling, live RTP delivery, DTLS-wrapped RTP protection, controlled impairment, and namespace-separated media paths.

10. Recommended Supervisor Discussion Points

Topic	Suggested Discussion
Novelty	Emphasize compact reference-based MCP/A2A coordination linked to secured media-plane validation.
Evaluation strength	Show progression from deterministic telemetry to semi-live, live RTP, DTLS-RTP, impairment, and namespaces.
Limitations	Current tests are single-machine; next validation should use two machines, Mininet, ns-3, or testbed.
Journal direction	Position Step 95 onward as agentic closed-loop media control under proof-governed trust decisions.
Figures	Use Step 87, Step 91, and Step 93 figures as core evaluation visuals.